FREE DIFFERENTIALS, REDUCEDNESS AND A PERFECT BASE FIELD FORCE SMOOTHNESS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter text immediately before the lemma supplies a counterexample showing that reducedness cannot simply be dropped in characteristic p, that is, the example Spec(k[t]/(t^p - a)) with free differentials which is not smooth; that remark lies outside the counted span.

Editorial note. The counted spans carry 8 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-char-p-differentials-free-smooth is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- A finitely generated field extension of a perfect field is separably generated. Algebra, Section algebra-section-separability, invoked at proof lines 4579-4581. Supplies that the module of differentials of the function field is free of rank the transcendence degree.
- The local ring of a reduced scheme at a generic point of a component is a field. Algebra, Lemma algebra-lemma-minimal-prime-reduced-ring, invoked at proof lines 4576-4578. Supplies the identification of the local ring at eta_i with kappa(eta_i).
- Dimension of a fibre at a point as a transcendence degree. Morphisms, Lemma morphisms-lemma-dimension-fibre-at-a-point, invoked at proof lines 4583-4585. Supplies dim_{eta_i}(X_i) = rank of the differentials at eta_i.
- Dimension at a point and the criterion for smoothness over a field by comparison with the rank of the differentials. Algebra, Lemmas algebra-lemma-dimension-at-a-point-finite-type-over-field and algebra-lemma-characterize-smooth-over-field, invoked at proof lines 4590-4593. Supplies the final identification of dim_x(X) with rank_x(Omega) and hence smoothness at x.

Source reference. Original lemma 00TT: sources/stacks-chow/tags.txt, source lines 1087–1087. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00OT: sources/stacks-chow/tags.txt, source lines 907–907. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 00EU: sources/stacks-chow/tags.txt, source lines 548–548. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 030I: sources/stacks-chow/tags.txt, source lines 3949–3949. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02FX: sources/stacks-chow/tags.txt, source lines 3208–3208. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01T6: sources/stacks-chow/tags.txt, source lines 2363–2363. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01OZ: sources/stacks-chow/tags.txt, source lines 2212–2212. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0052: sources/stacks-chow/tags.txt, source lines 194–194. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. Assume*

(1) *X is locally of finite type over k,*
(2) *$\Omega_{X/k}$ is locally free,*
(3) *X is reduced, and*
(4) *k is perfect.*

Then the structure morphism $X \to \mathrm{Spec}(k)$ is smooth.

Proof. Let $x \in X$ be a point. As X is locally Noetherian (see Morphisms, Lemma 01T6) there are finitely many irreducible components $X_1, \ldots, X_n$ passing through x (see Properties, Lemma 01OZ and Topology, Lemma 0052). Let $\eta_i \in X_i$ be the generic point. As X is reduced we have $\mathcal{O}_{X,\eta_i} = \kappa(\eta_i)$, see Algebra, Lemma 00EU. Moreover, $\kappa(\eta_i)$ is a finitely generated field extension of the perfect field k hence separably generated over k (see Algebra, Section 030I). It follows that $\Omega_{X/k,\eta_i} = \Omega_{\kappa(\eta_i)/k}$ is free of rank the transcendence degree of $\kappa(\eta_i)$ over k. By Morphisms, Lemma 02FX we conclude that $\dim_{\eta_i}(X_i) = \mathrm{rank}_{\eta_i}(\Omega_{X/k})$. Since $x \in X_1 \cap \ldots \cap X_n$ we see that

$$\mathrm{rank}_x(\Omega_{X/k}) = \mathrm{rank}_{\eta_i}(\Omega_{X/k}) = \dim(X_i).$$

Therefore $\dim_x(X) = \mathrm{rank}_x(\Omega_{X/k})$, see Algebra, Lemma 00OT. It follows that $X \to \mathrm{Spec}(k)$ is smooth at x for example by Algebra, Lemma 00TT. □